

Klokd

MVP Specification · Definition of Done · Sprint Backlog

Product B · Casual Labour Marketplace

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Classification: Confidential · Internal Use Only

Status: Living Document — update as Phase 0 progresses

23 Sprints · 46 Weeks · 387 Story Points · Controlled Beta Launch

1. Executive Summary

Klokd is a two-sided M-Pesa Native mobile platform connecting verified casual workers with businesses in hospitality, health, and retail. Workers find shifts near them in seconds. Employers post in under 2 minutes. M-Pesa pays workers within 30 minutes of clock-out. Every shift generates a compliant employment contract. The Compliance Engine enforces WIBA, minimum wage, Section 37 thresholds, and statutory deduction calculations automatically.

Platform identity: Klokd is a platform company, not an app. The Worker App and Employer App are Layer 2 clients. The Layer 1 API is the business — callable by any future authenticated client including the Year 2 EOR portal.

Beachhead: Nairobi hospitality sector. 50 employers, 200 workers at controlled beta.

2. Architecture Overview

Three-Layer Model

Layer	Name	Technology	Key Services
Layer 1	Core Platform API	Node.js / Express 5 / TypeScript AWS af-south-1 / PostgreSQL RDS	Shift · Payment · Compliance Engine Contract · Identity · Notifications Webhook · Analytics Event Log
Layer 2	Products	React Native (Android-first) React web (Admin Portal)	Worker App · Employer App Admin Portal · EOR Portal (Year 2)
Layer 3	Ecosystem	Third-party integrations	Safaricom Daraja B2C · WIBA insurer KRA / iTax · Enterprise HR (Year 2)

Key Architectural Decisions (Locked)

Decision	Detail
REST API for MVP	GraphQL deferred to Year 2
Multi-tenant schema	tenant_id on all tables from Sprint 4
Event-driven lifecycle	ShiftEventLog: posted → matched → confirmed → active → completed → disputed → paid
Payment decoupled	PaymentService has zero direct dependency on ShiftService
Compliance Engine	Discrete Layer 1 service · All statutory rates in config, never hardcoded
AHL toggle	On/off via config without code deployment (legal status unresolved)
Analytics	geo_hash not raw GPS · DPA 2019 data minimisation
AWS af-south-1	Kenyan data residency requirement
Manual matching MVP	Filtered list by proximity, rating, availability — no algorithm
GPS clock-in	500m radius · Offline queue for low connectivity
Escrow model	Employer funds held from confirmation · Auto-release 4 hours post clock-out
Android-first	Worker persona device reality

3. MVP Scope

In Scope — Beta (May/June 2026)

Worker App (5-screen onboarding + 10 functional screens)

- Welcome carousel → Privacy & Consent → Verify ID → Skills & Certs → M-Pesa Setup (5 onboarding screens)
- Home: Stats card + available shift feed (proximity-sorted)
- Shift Detail: Full info, employer profile, payment guarantee, contract preview, Accept/Decline
- Contract Acknowledgement: Auto-generated contract, explicit acceptance, PDF download
- Active Shift — Clock In: GPS radius (500m), WIBA gate, offline queue
- Active Shift — In Progress: Timer, employer contact, Clock Out CTA
- Payment Confirmed: M-Pesa celebration, deduction breakdown, pay statement download, rating prompt
- Shifts Tab: Upcoming / Active / History with 7-year retention enforced
- Pay Tab: Monthly aggregation — gross, PAYE, NSSF, SHIF, AHL (if active), net to M-Pesa
- Profile/Me Tab: Portable reputation, Privacy & Data panel (data subject rights)
- Dispute Flow: Initiate, describe, evidence, submit, status tracking

Employer App (4-screen onboarding + 10 functional screens)

- Welcome → Business Verify (KRA PIN) → WIBA & Insurance → M-Pesa Setup (4 onboarding screens)
- Dashboard: Confirmed/pending shift status, Post a Shift CTA
- Post a Shift: Role / date / time / rate with minimum wage gate + Section 37 alert + rate intelligence
- Select Worker: Ranked cards (proximity + rating), show-up rate prominent
- Active Shift Monitoring: Worker clocked in confirmation, elapsed timer, dispute button
- Clock-Out Confirm: Release payment or dispute, 4-hour auto-release countdown
- Shifts Tab: All shifts — completed, active, pending, disputed
- Team Tab: Saved workers (favourites + one-tap re-hire), blocked workers
- Pay/Billing Tab: Escrow balance, per-shift spend, transaction history, CSV export
- Rating Screen: 1–5 stars + optional text, post-shift
- Dispute Flow: No-show / incomplete / conduct · Evidence upload · Status tracking

Admin Portal (4 panels)

- Compliance Configuration: Statutory rate toggles with audit trail
- Data Subject Request Management: Request log, response tracking, 72-hour ODPC notification workflow
- Breach Response Panel: Detection log, ODPC notification timer, pre-filled template, status tracking
- Monthly Compliance Dashboard: PAYE/NSSF/SHIF aggregations, KRA iTax export

Compliance Engine — Layer 1 (Sprint 7)

- POST /compliance/calculate — PAYE, NSSF, SHIF, AHL per shift and monthly aggregate
- GET /compliance/wiba/:shiftId — Hard gate before clock-in (confirmed or blocked)
- GET /compliance/section37/:workerEmployerPair — Alert at 20/25/30-day thresholds
- POST /compliance/minwage/validate — Block shift posting below applicable minimum
- POST /compliance/contract/generate — Employment Act s.9-compliant contract

Out of Scope — MVP

Feature	Rationale
Algorithm matching	Manual sufficient at 50/200 beta cohort
GraphQL API	REST sufficient; adds complexity before partner demand
EOR Portal	Pending legal opinion on employment classification
Enterprise HR API	Layer 3, Year 2
Per-shift micro-WIBA	Requires insurance agreement; Option A (employer policy) for MVP
Labour Market Intelligence API	Year 3 revenue stream
iOS app	Android-first; iOS in Year 1 post-beta
AHL deductions	Config-off pending legal opinion on enforceability
MFI micro-loan integration	Year 2
Pan-Africa white-label	Year 4+
Product A (HR management)	Deferred indefinitely

4. Definition of Done

Story-Level DoD

- All acceptance criteria pass
- Unit tests written and passing (coverage $\geq 80\%$ for service layer)
- Integration test written and passing
- No TypeScript errors (tsc --noEmit clean)
- ESLint passes with zero warnings
- API endpoint documented (if applicable)
- Error states handled — no unhandled promise rejections
- Mobile screens reviewed at 360px and 390px viewport widths
- Accessibility: touch targets $\geq 44\text{px}$, sufficient colour contrast
- Sensitive data (National ID, M-Pesa number, GPS) handled per DPA 2019
- Story status updated to Done in Trello

Sprint-Level DoD

- All committed stories meet story-level DoD
- Sprint review conducted — both founders present
- All API endpoints deployed to staging environment
- Staging smoke test passing (Postman collection executed)
- No P0 or P1 bugs open
- Sprint retrospective completed, action items logged
- Reboot Pack updated with any new decisions
- Risk register reviewed — any triggered risks escalated

Phase-Level DoD

- All sprints in phase meet sprint-level DoD
- Phase deliverables demonstrated to at least one external stakeholder
- Legal/compliance dependencies for next phase confirmed
- Infrastructure costs reviewed against budget
- Investor update prepared and sent if phase is a major milestone

MVP DoD — Beta-Ready

- All Phase 0–7 sprints meet sprint-level DoD
- 50 employers + 200 workers onboarded and active
- End-to-end shift flow tested: post $\rightarrow$ match $\rightarrow$ clock-in $\rightarrow$ clock-out $\rightarrow$ M-Pesa $\rightarrow$ rating
- WIBA hard gate confirmed blocking clock-in on uncovered shifts
- Minimum wage gate confirmed blocking below-minimum postings
- Section 37 alert triggering correctly at threshold

- Daraja B2C production: real KES disbursed to real M-Pesa number
- Escrow auto-release confirmed at 4 hours post clock-out
- ODPC registered as Data Controller and Data Processor
- Privacy Policy published · DPAs executed with all third-party processors
- All data confirmed in AWS af-south-1 only
- API P95 response time < 500ms under 80 concurrent users
- Load test: 80 concurrent users, 30 minutes, zero errors
- KRA PIN obtained · PAYE filing setup on iTax
- NSSF and SHIF registrations complete

5. Sprint Backlog

23 sprints · 46 weeks · 387 story points · 2-week sprint cadence · Team: Chamia (CPO/PO) + 2 developers + part-time designer (from Sprint 11)

Phase	Sprints	Weeks	Points	Key Deliverable
Phase 0 — Validation	S1–S2	1–4	41	Legal opinion · 50 interviews · Infrastructure
Phase 1 — Core API	S3–S6	5–12	78	Auth · Shift · Match · GPS · Notifications
Phase 2 — Compliance	S7	13–14	26	Compliance Engine · Contracts · Webhooks
Phase 3 — Worker App	S8–S11	15–22	69	Complete worker flow end-to-end
Phase 4 — Employer App	S12–S15	23–30	62	Complete employer flow end-to-end
Phase 5 — Payment	S16–S18	31–36	42	Daraja production · Escrow · Pay statements
Phase 6 — Admin	S19–S21	37–42	35	Compliance config · DPA rights · Security
Phase 7 — Beta	S22–S23	43–46	34	First live shift · Go-live
TOTAL	23 sprints	46 weeks	387	Controlled beta launch

Phase 0

Validation & Legal Foundation

Weeks 1–4

Confirm market demand · Secure legal opinion · Stand up infrastructure · Recruit beta cohort

Exit criteria: Legal opinion received · 20 employer + 30 worker interviews complete · All P0 compliance actions initiated

Sprint 1

Weeks 1–2

19 pts

Objective: Legal counsel engaged · Infrastructure stood up · Investor outreach initiated

ID	User Story	Pts	Acceptance Criteria
S1-01	Engage Kenyan labour counsel for written legal opinion on PAYE/NSSF/WIBA/entity classification	2	SoW signed · Briefing complete · Opinion deadline agreed (end Week 4)
S1-02	Stand up AWS dev/staging/production environments in af-south-1	3	Three environments created · RDS PostgreSQL provisioned · S3 buckets with encryption at rest
S1-03	Complete Daraja B2C sandbox integration including all failure scenarios	3	Sandbox credentials active · B2C initiation succeeds · All failure scenarios documented
S1-04	Submit Safaricom production access application	1	Application submitted · Reference number filed
S1-05	Finalise and pilot-test employer interview script	2	Covers staffing pain, WA reliance, payment timing, WTP · Pilot-tested with 1 contact
S1-06	Finalise and pilot-test worker interview script	2	Covers income reliability, ID concerns, M-Pesa trust, shift discovery
S1-07	Send Wave 1 investor outreach to 10+ targets	2	Pack final · Savannah Fund, Kepple, Safaricom Spark Fund contacted
S1-08	Submit ODPC Data Controller and Data Processor registration applications	2	Both applications submitted · Reference numbers filed
S1-09	Submit KRA PIN application	1	Application submitted · Reference filed
S1-10	Submit BRS company registration	1	Application submitted · Certificate expected within 7 days

Sprint 2

Weeks 3–4

22 pts

Objective: 20 employer + 30 worker interviews complete · Legal opinion received

ID	User Story	Pts	Acceptance Criteria
S2-01	Complete 20 employer in-person interviews in Nairobi hospitality	5	20 interviews complete · Findings documented · ≥10 express intent to use
S2-02	Complete 30 worker in-person interviews	5	30 interviews complete · Findings documented · ≥20 express intent to join
S2-03	Receive and review legal opinion	3	Written opinion received · PAYE/NSSF/WIBA positions confirmed · AHL status confirmed
S2-04	Confirm WIBA coverage option (A or B) and finalise clock-in gate architecture	2	Option A (employer-held policy) confirmed for MVP · WIBA declaration flow designed
S2-05	Design and review base database schema	3	ERD approved · All core tables with tenant_id · ShiftEventLog designed
S2-06	Confirm AHL enforceability and set toggle	2	Legal counsel confirms status · Toggle set · Documented in Reboot Pack
S2-07	Obtain minimum wage schedules for hospitality, health, retail	2	Current schedules obtained · Staged in config table (not hardcoded)

Phase 1**Core Platform API**

Weeks 5–12

All Layer 1 services scaffolded and tested. Database live. Authentication working.

Exit criteria: Postman collection for all core endpoints passing on staging · No P0 bugs**Sprint 3**
Weeks 5–6**19 pts***Objective: Project scaffold · Authentication · Identity Service · Database live*

ID	User Story	Pts	Acceptance Criteria
S3-01	Complete project scaffold — Express 5/TypeScript monorepo, CI/CD pipeline	3	ESLint + Jest configured · GitHub branch protection · GitHub Actions → AWS staging pipeline live
S3-02	Implement JWT authentication with OTP via Africa's Talking	5	JWT + refresh tokens · Role-based: worker/employer/admin · OTP expires 5min · Rate limit: 3 attempts/10min
S3-03	Build Identity Service — National ID verification and selfie liveness	5	ID images encrypted in S3 · Verification status: pending/approved/rejected · ID number hashed (not plaintext)
S3-04	Implement append-only audit log on all sensitive operations	3	audit_log table: user_id, action, timestamp, ip_hash · Covers identity, payment, admin config · No deletes
S3-05	Implement ShiftEventLog for all shift state transitions	3	shift_events: shift_id, from_state, to_state, actor_id, timestamp, metadata · All transitions logged

Sprint 4
Weeks 7–8**21 pts***Objective: Shift Service · Employer posting · Minimum wage gate · Worker profile*

ID	User Story	Pts	Acceptance Criteria
S4-01	Build Shift Service — POST /shifts with full validation	5	Role, date, time, rate, location validated · Returns shift_id · Status: posted
S4-02	Implement minimum wage validation on shift creation	3	Rate vs config table · Below minimum: 422 with minimum + sector · Updatable without code deploy
S4-03	Build worker profile endpoint — skills, certifications, availability	3	Profile completeness score · Incomplete profiles cannot apply for shifts
S4-04	Build employer profile endpoint — business + WIBA declaration	3	KRA PIN, contact, WIBA policy reference + insurer + expiry · WIBA mandatory before first shift
S4-05	Build GET /shifts/available — proximity-filtered shift feed	5	Filter: lat/lng radius (default 5km), role, date · Sorted distance ASC · Employer rating included
S4-06	Build shift application endpoint — POST /shifts/:id/apply	2	Worker must be identity-verified · No overlapping confirmed shifts · Status: pending

Sprint 5
Weeks 9–10

19 pts

Objective: Worker matching · Employer selection · Shift confirmation · Escrow initiation

ID	User Story	Pts	Acceptance Criteria
S5-01	Build GET /shifts/:id/applicants — ranked worker list	5	Sorted: distance ASC, rating DESC, show-up rate DESC · All worker card data included
S5-02	Build shift confirmation — employer selects worker	3	Shift status: confirmed · Worker + other applicants notified · ShiftEventLog updated
S5-03	Build worker acceptance and contract generation trigger	3	Shift status: accepted · Contract Service triggered · Employer notified
S5-04	Implement escrow funding via STK Push on shift confirmation	5	STK Push initiated · Funding confirmed: escrow_status = funded · Declined: shift reverts to posted
S5-05	Implement Section 37 threshold monitoring and alerts	3	Alert at 20 days (warning) · 25 days (explicit acknowledgement) · 30 days (hard block)

Sprint 6
Weeks 11–12

19 pts

Objective: GPS clock-in · WIBA gate · Clock-out · Notifications · Dispute initiation

ID	User Story	Pts	Acceptance Criteria
S6-01	Build GPS clock-in with 500m validation and offline queue	5	500m radius enforced · Offline: queued and replayed · GPS hashed (geo_hash) before storage
S6-02	Implement WIBA hard gate before every clock-in	3	GET /compliance/wiba/:shiftId called · Not confirmed: 403 with reason · Surfaced to both apps
S6-03	Build clock-out and auto-release timer	3	Clock-out recorded · Status: completed · Auto-release timer started (4 hours) · ShiftEventLog updated
S6-04	Implement FCM notifications with WhatsApp Business API fallback	5	FCM primary · WA fallback at 5min · All events logged · Bilingual templates · Tested on device
S6-05	Build dispute initiation endpoint	3	Dispute types: no-show/incomplete/conduct/payment · Auto-release paused · Both parties + admin notified

Phase 2

Compliance Engine & Contracts

Weeks 13–14

All statutory compliance logic live as discrete Layer 1 services with clean API surface.

Exit criteria: All compliance endpoints tested · WIBA gate integration tested · Contract generation producing Employment Act s.9-compliant output

Sprint 7

Weeks 13–14

26 pts

Objective: Compliance Engine · Contract Service · Webhook Service · Analytics Event Log

ID	User Story	Pts	Acceptance Criteria
S7-01	Implement PAYE calculation — per shift and monthly aggregate	5	2025/26 bands from config · Personal relief applied · All rates in config (never hardcoded)
S7-02	Implement NSSF Tier I and Tier II calculation with monthly aggregation	3	Tier I: 6% up to KES 480 · Tier II: 6% on KES 8,001–72,000 · Ceiling enforced
S7-03	Implement SHIF calculation with monthly aggregation	3	2.75% of gross monthly · Minimum KES 300 · Returns shift_deduction, month_gross
S7-04	Implement AHL toggle — config-driven, no code deploy required	2	ahl_enabled in compliance_config · Toggle change writes to audit_log · Zero deduction when disabled
S7-05	Implement minimum wage validation endpoint	2	POST /compliance/minwage/validate · Config table · Updatable without deploy
S7-06	Build Contract Service — Employment Act s.9-compliant contract per shift	5	PDF generated · All required fields per s.9 · Stored in S3 (7-year retention) · Digital acceptance recorded
S7-07	Build Webhook Service with HMAC-SHA256 signing and retry logic	3	WebhookSubscription model · 3-attempt retry with backoff · WebhookDeliveryLog · Events: shift/payment/dispute
S7-08	Implement Analytics Event Log — geo_hash, no PII	2	All events logged · geo_hash not raw GPS · No raw PII · Indefinite retention (aggregated)
S7-09	Implement NITA levy calculation	1	KES 50/worker/month · Monthly aggregation enforced · Included in compliance calculation response

Phases 3–7 — Backlog Summary

Full user story detail for Phases 3–7 is available in the interactive Sprint Backlog artifact. The following table summarises all sprints by phase with objectives and acceptance criteria counts.

Sprint	Weeks	Pts	Sprint Objective
S8	15–16	21	Worker onboarding — all 5 screens · Privacy/consent · ID verify · Skills · M-Pesa
S9	17–18	18	Home screen · Shift Detail · Contract acknowledgement · Accept/Decline · Notifications
S10	19–20	16	Clock-In (GPS + WIBA gate) · Active Shift · Clock-Out · Payment Confirmed
S11	21–22	14	Shifts Tab · Pay Tab (deduction breakdown) · Profile/Me Tab · Dispute Flow
S12	23–24	18	Employer onboarding — all 4 screens · Business verify · WIBA capture · M-Pesa setup
S13	25–26	19	Dashboard · Post a Shift (min wage gate + S37 alert) · Select Worker
S14	27–28	14	Active monitoring · Clock-Out confirm · Rating screen · Dispute Flow
S15	29–30	11	Shifts Tab · Team Tab (favourites + re-hire) · Pay/Billing Tab
S16	31–32	18	Daraja B2C production · Escrow STK Push · Deduction-adjusted disbursement
S17	33–34	11	Pay statement PDF generation · 7-year retention enforcement · Auto-release
S18	35–36	13	Payment failure handling · Dispute resolution · Monthly reconciliation
S19	37–38	13	Compliance Config panel · Monthly dashboard · Worker/employer management
S20	39–40	13	Data Subject Rights workflow · Breach Response Panel · DPA rights in-app
S21	41–42	9	API monitoring · Encryption audit · Rate limiting · OWASP review
S22	43–44	18	E2E integration testing · Load test 80 users · Beta cohort onboarded
S23	45–46	16	First live shift · Real KES disbursed · P0/P1 bugs resolved · Investor update

6. Compliance Dependencies — Critical Path

The following actions must complete before the sprint that depends on them. Missing these blocks development. No exceptions.

Action	Owner	Deadline	Blocks
Legal opinion: PAYE/NSSF/WIBA/entity classification	Chamia	End Sprint 1	Everything
ODPC registration confirmed	Chamia	End Sprint 2	Sprint 8 (identity collection)
KRA PIN obtained	Chamia	End Sprint 1	Sprint 19 (PAYE remittance)
WIBA option confirmed (A or B)	Ivy	End Sprint 2	Sprint 6 (clock-in gate)
AHL enforceability confirmed	Legal counsel	End Sprint 2	Sprint 7 (Compliance Engine)
Escrow model / CBK licensing position	Legal counsel	End Sprint 2	Sprint 16 (Payment)
Minimum wage schedules obtained	Developer	End Phase 1	Sprint 7 (min wage gate)
DPAs with AWS, Safaricom, WhatsApp, Africa's Talking	Chamia	Before Sprint 22	Go-live
Daraja production keys	Chamia	Before Sprint 16	Payment Service
DPIA completed	Ivy + Legal	Before Sprint 8	Worker onboarding
Privacy Policy published	Legal	Before Sprint 22	DPA compliance

7. Technology Stack

Layer	Technology
Mobile apps	React Native · Android-first · Expo (SDK pinned)
Backend framework	Express 5 / TypeScript
Runtime	Node.js 22 (Railway) · Node.js 16 (local macOS Catalina)
Database	PostgreSQL (AWS RDS, af-south-1)
ORM	Prisma
File storage	AWS S3 (af-south-1, encryption at rest, 7-year lifecycle)
Authentication	JWT + refresh tokens · OTP via Africa's Talking
Push notifications	FCM (primary) · WhatsApp Business API (fallback at 5min)
Payments	Safaricom Daraja B2C
Hosting	Railway (backend) · AWS af-south-1
DNS / Proxy	Cloudflare
Monitoring	CloudWatch · PagerDuty

Layer	Technology
CI/CD	GitHub Actions
Testing	Jest (unit/integration) · k6 (load) · Postman (API)

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